MDCD-DB: Benchmarking Multimodal Deepfake Continual Detection Against Evolving Forgery Threats

A Baselines Description

We evaluate the proposed MDCD benchmark with implemented continual learning baselines to establish benchmark results. 1) Finetune trains on each incremental dataset directly. 2) Replay-based methods [2, 4, 7] select a limited original videos in memory buffer for retraining in various strategies. 3) Regularization-based methods [1–3, 7, 8] introduce regularization items to prevent the old knowledge from forgetting. 4) AL-based methods [6, 9, 10] leverage the analytic learning paradigm to update the weights instead of back-propagation algorithms. The technique and details they adopted are illustrated in Table 5, which are primarily divided into four types. Exemplar represents the method that preserves a certain number of exemplars for replay. Distillation involves adding a constraint term to the model update. Prototype denotes the method that retains old knowledge by calculating a class prototype. Analytic indicates that the method leverages analytic learning algorithms to update weights of the classification head. Each baseline is described in detail as follows:

- Finetune: training on each incremental dataset directly. This method easily leads to catastrophic forgetting of historical deepfake knowledge due to domain gaps.
- Replay: randomly preserves some examples from each trained dataset and re-trains with them in the subsequent tasks. It mitigates catastrophic forgetting by preserving previous training samples.
- iCaRL (CVPR 2017): selects the samples closest to the class mean at each phase and replays them in the following.
- LwF (TPAMI 2018): uses only new task data to train the network while preserving the original capabilities by introducing a regularization loss between the current network and the old network.
- PASS (CVPR 2021): augments the embedding of the prototype of the old class in the feature space to retain the old decision boundary, and reduce task-level overfitting through self-supervised learning. It is an exemplar-free method without extra storage cost.
- ACIL (NeurIPS 2022): trains the feature extractor in the base training phase and updates the analytic classifier with the frozen feature extractor in incremental learning phases. ACIL is an AL-based paradigm using least squares to calculate the closed-form solution of the classification weight. It achieves a comparable performance to joint learning on all seen datasets.
- Fetril (WACV 2023): generates pseudo-features of old classes by simply translating the features of new classes with the stored old class mean. It only updates the classifier rather than the whole network, gaining a lower time cost.
- DFIL (MM 2023): selects central and hard representative samples for replay to cope with different deepfake data distributions. It regularizes the model in both feature-level and label-level to mitigate catastrophic forgetting. DFIL is the first domain-incremental learning framework for deepfake detection, and it is a model-agnostic method that easily adapts to MDCD.
- DS-AL (AAAI 2024): introduces a compensation stream improving the inherent under-fitting limitation due to adopting linear mapping based on ACIL.
- MMAL (MM 2024): is a multimodal analytic learning framework introducing a modality fusion module that harnesses individual knowledge from audio and visual modality in tandem. MMAL leverages fine-grained modality knowledge to alleviate the under-fitting limitations.
- TagFex (CVPR 2025): provides additional task-agnostic features for subsequent tasks and integrates the task-agnostic features with the task-specific features. It helps capture diverse features to combat the feature collision.

Note that the memory buffer size for replay-based continual baselines varies from different incremental settings. We set the number of exemplars per class to 500, 100, and 1000 for the sequence incremental setting, cross-technology incremental setting, and cross-modality setting, respectively. The critical hyperparameters of each baseline are referenced from the original algorithm, unless specifically stated.

B Implementation Details

We adopt the multimodal deepfake detection model [5] constructed by a heterogeneous graph attention network as our feature extractor. It can efficiently model intra- and inter-modal relationships of audio-visual data both at spatial and temporal scales. This model has made remarkable detection capacity on the FakeAVCeleb and the LAV-DF dataset, providing a reliable basis for MDCD. Following [5], we extract four frames with a fixed offset from every 10 of the 100 frames in each input video to form a visual stream, resulting in a total of 40 frames. The input face frames are resized to 128×128. All models are trained for five epochs on the training data during incremental learning phases, except for forward-only methods that train five epochs during base training and train one-epoch in the incremental learning phases. We use the Adam optimizer with a learning rate of 0.001 and betas of 0.9 and 0.999. We set $N = 128$ and $\gamma = 0.03$.

Note that the memory buffer size for replay-based continual baselines varies from different incremental settings. We set the number of exemplars per class to 5% of the current training dataset, with a maximum of 500. The critical hyperparameters of each baseline are referenced from the original algorithm, unless specifically stated.

C Evaluation Metrics

Following the widely used metrics, we adopt accuracy ($ACC_{k,i}$) and ROC Curve ($AUC_{k,i}$) to quantify the deepfake detection performance of the i-th task after training at phase k. To present the overall performance during continual learning, we adopt the average accuracy $AA_k = \frac{1}{k}\sum_{i=1}^{k} ACC_{k,i}$ and the average AUC $AAUC_k =$

Table 5: The technique and description of baselines used.

Method	Technique	Description
Finetune	-	Direct training on incremental datasets causes catastrophic forgetting of historical deepfake knowledge due to domain gaps.
Replay (Rondom)	Exemplar	Randomly preserving samples from previous datasets and retraining them in subsequent tasks to mitigate catastrophic forgetting.
iCaRL (CVPR 2017)	Exemplar	Selects the samples closest to the class mean at each phase and replays them in the following.
LwF (TPAMI 2018)	Distillation	A regularization loss constraining current and old networks maintains original capabilities with new task data only.
PASS (CVPR 2021)	Distillation & Prototype	Prototype embedding augmentation retains old decision boundaries and reduces task-level overfitting via self-supervised learning, exemplar-free without storage cost.
ACIL (NeurIPS 2022)	Analytic	ACIL trains the feature extractor initially and updates the analytic classifier with frozen features via least squares, achieving joint-learning comparable performance without retraining.
Fetril (WACV 2023)	Prototype	Pseudo-feature generation via new-class translation with stored old means updates only the classifier, achieving lower time costs.
DFIL (MM 2023)	Exemplar & Distillation	DFIL selects central and hard exemplars for replay, employing feature and label regularization to mitigate catastrophic forgetting. As the first domain-incremental framework for deepfake detection, this model-agnostic method adapts seamlessly to MDCD.
DS-AL (AAAI 2024)	Analytic	Introduces a compensation stream that addresses the inherent under-fitting limitation of ACIL from linear mapping.
MMAL (MM 2024)	Analytic	MMAL's modality fusion module harnesses fine-grained audio-visual knowledge to alleviate under-fitting limitations.
TagFex (CVPR 2025)	Exemplar & Distillation	Task-agnostic feature integration with task-specific features captures diverse representations to combat feature collision.

$\frac{1}{k}\sum_{i=1}^{k} AUC_{k,i}$ that denote the test performance on all previous tasks. As for the incremental settings without task identity during testing, let ACC_k and AUC_k indicate the test accuracy on all seen datasets $\mathbf{D}_{1:k}$. After all continual learning tasks, We employ $AIA = \frac{1}{N}\sum_{k=1}^{N} AA_k$ and $AIAUC = \frac{1}{N}\sum_{k=1}^{N} AAUC_k$ to evaluate the final performance on all seen tasks. Another metric is Last ACC defined as $Last\ ACC = AA_N$, which denotes the final average accuracy on all phases.

The Backward Transfer (BWT) rate is used to evaluate the extent of forgetting old knowledge by the new model, which is the proportion of the decrease in prediction accuracy for previous tasks. It is calculated using $BWT_k = \frac{1}{k}\sum_{i=1}^{k-1}(ACC_{k,i} - ACC_{i,i})$ and $BWT = \frac{1}{N-1}\sum_{k=2}^{N} BWT_k$. Similarly, the Average Forgetting (AF) rate is the proportion of the decrease from the best performance in ACC. In detail, $AF = \frac{1}{N-1}\sum_{k=2}^{N} F_k$ and $F_k = \frac{1}{k-1}\sum_{i=1}^{k-1}\max_{\tau\in\{1,\ldots,k\}}(ACC_{\tau,i} - ACC_{k,i})$. For the incremental settings without task identity, AF is refined as $AF = \frac{1}{N-1}\sum_{k=2}^{N}\max_{\tau\in\{1,\ldots,k\}}(ACC_{\tau} - ACC_k)$.

We introduce Forward Transfer (FWT) to evaluate learning plasticity, which measures the impact of learned tasks on the current task. $FWT_k = \frac{1}{k-1}\sum_{i=2}^{k}(ACC_{i,i} - \widetilde{ACC_i})$, where $\widetilde{ACC_i}$ is the classification accuracy of a randomly-initialized reference model trained with D_i for the i-th task. Similarly, we use FWT to evaluate the average learning plasticity as $FWT = \frac{1}{N-1}\sum_{k=2}^{N} FWT_k$.

D Definition of Experimental Settings

D.1 Sequence Incremental Learning

Existing multimodal deepfake datasets are released publicly in sequence, limiting the ability to train a detection model offline with all data at one time. To adapt to successive multimodal deepfake data, we split the dataset in time order into six batches for incremental learning. An adaptive and effective multimodal deepfake detector should be capable of continually learning from emerging deepfake data while maintaining strong performance on previously encountered samples. We consider two critical scenarios: task incremental learning (TIL) and domain incremental learning (DIL). The forward sequence is {DFDC-P (2019), DFDC (2020), KoDF (2021), FakeAVCeleb (2021), LAV-DF (2022), AV-Deepfake1M (2023)}.

Task Incremental Learning. Each newly released dataset is treated as a separate incremental learning task. MDCD methods try to learn from the new dataset while not forgetting the historical deepfake knowledge. Therefore, task identities are provided in training and testing to evaluate the performance of each previous task. We evaluate the performance of each previous task from $\mathbf{D}_1$ to $\mathbf{D}_k$ at the k-th phase. Besides, to evaluate the robustness of MDCD methods to the order of sequential incremental tasks, MDCD-DB reports benchmark results on the reverse sequence {AV-Deepfake1M, LAV-DF, FakeAVCeleb, KoDF, DFDC, DFDC-P}.

Domain Incremental Learning. Since the source of the test video in the real-world may be unknown to detectors, task identities are not provided during testing. Thus, we evaluate the union test sets from all seen datasets $\mathbf{D}_{1:k}$ after each phase *Domain k*.

D.2 Cross-Technology Incremental Setting

With the maturity of face and audio synthetic technologies, new multimodal deepfake technologies have been emerging. In addition to detecting whether they are deepfake videos, it is also necessary to identify which type of deepfake method they were generated by. To evaluate the continual learning performance of cross-technology scenarios, we construct a sequential multimodal deepfake video set according to the release time of the deepfake technology {Faceswap, RTVC, FSGAN, Wav2lip, Faceswap-wav2lip, FSGAN-wav2lip}. To identify the specific deepfake technology, each deepfake technology should be classified by a unique class label $\mathbf{Y}_k \in \{1, 2, 3, 4, 5, 6\}$ in the class incremental learning.

D.3 Modality Incremental Setting

Deepfake technologies show a trend of gradually evolving from unimodal to multimodal. Meanwhile, deepfake detectors are required to adapt to various deepfake modalities. To this end, we divide the data into four categories: {real, video-only, audio-only

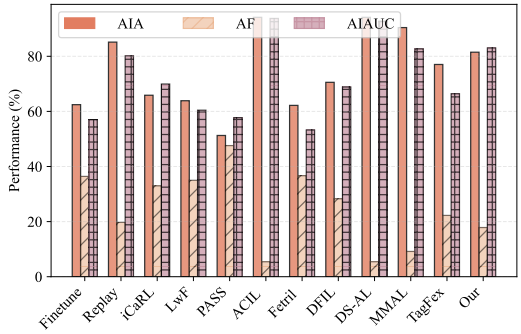

Figure 5: The overall performance under the domain incremental setting following a reverse sequence.

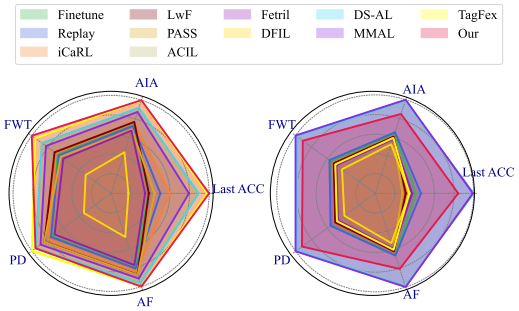

Figure 6: The left panel and the right panel indicate the overall performance under the cross-technology incremental setting in CIL following forward and reverse orders, respectively.

and video-audio}. They represent real samples, samples with only the visual modality forged, samples with only the audio modality modified, and samples where both the visual and audio modalities are forged, respectively.

E Additional Experiments

In order to promote further research in the MDCD scenario, we provide detailed experimental details for thorough evaluation. Tables 6 and 7 respectively detail the performance of each stage in various tasks under the sequential incremental scenario in both forward and reverse orders. We find an unavoidable forgetting for every baseline. Even the most effective methods can preserve forged knowledge well in some tasks, but they still struggle to address the feature differences between multimodal deepfake data, especially for Task 2 in the forward sequence.

AL-based methods demonstrate outstanding stability. From Table 2, ACIL and DS-AL drop less than 2% average accuracy for each task under the forward sequence incremental setting, demonstrating the best stability, followed by MMAL. Especially in the reverse sequence incremental setting in Table 2, Figure 5, AL-based approaches achieve almost zero in terms of *AF* from Table 2. They demonstrate a very slow downward trend in Figure 5. ACIL, DS-AL, and MMAL even achieve higher AIA and AIAUC than replay-based methods with lower average forgetting rate as illustrated in Figure 5. They have learned a general representation from a large-scale dataset in Task6 or Domain6, which is capable of extract discriminative features in the subsequent tasks. The stability originates from the strategy that preserves old multimodal deepfake knowledge by freezing the feature extractor and updating the classification weight in a closed-form instead of backpropagation algorithms. However, there exists a different situation that occurs in DIL following a forward training sequence from Table 3. Because analytic learning paradigms restrict their plasticity to emerging multimodal deepfake data. Learning few discriminative features from a large-scale dataset in Domain6 results in low accuracy.

Figure 6 demonstrates the overall performance (AIA, Last ACC, AF, BWT, and FWT) under the cross-technology incremental setting in CIL, and the left and right panels follow forward and reverse orders, respectively. For the forward order, it is observed that our method achieves the largest recognition area and superior performance across all evaluation metrics compared to other baseline methods, followed by TagFex in second place, respectively. Notably, the optimal and sub-optimal methods are replay-based strategies. In contrast, under the reverse incremental order, MMAL performs the best, with DS-AL and ACIL ranking second and third. These leading methods are all based on analytic learning frameworks. The primary reason for this phenomenon is that as forgery techniques evolve incrementally over time, their difficulty and complexity increase. Consequently, in the reverse incremental learning scenario, AL-based methods are exposed to more complex forgery knowledge earlier in the process. Combined with their inherent capability for absolute memorization, this early exposure to harder samples results in stronger overall performance in the reverse order.

Table 9 illustrates the detailed performance of each phase in various tasks in the modality incremental scenario. Finetune and PASS suffer 33.63% and 34.82% accuracy drops on video-only deepfake videos after training on audio-only deepfake data, while others retain video deepfake knowledge. It demonstrates that the deepfake artifact features between the visual and audio unimodal are relatively simple and direct transformations. Nevertheless, most baselines encounter catastrophic forgetting after training on video-audio deepfakes. Especially, AL-based methods still drop over 16% accuracy, though freezing the feature extractor. thereby verifying a more complex feature drift from unimodal deepfakes to multimodal deepfakes. Our method surpasses other baselines attributed to the critical memory and domain-invariant frequency alignment strategies to reduce the cross-modality gap.

References

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Table 6: Detailed performance at each phase in the forward sequence incremental setting.

Method	Datasets	Task1		Task2		Task3		Task4		Task5		Task6		AAUC ↑	Fk ↓
		ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC		
Finetune	Task1	89.94	78.46	-	-	-	-	-	-	-	-	-	-	89.94	-
	Task2	72.85	74.18	83.94	84.83	-	-	-	-	-	-	-	-	78.40	17.09
	Task3	49.22	59.19	54.21	58.13	99.81	99.69	-	-	-	-	-	-	67.75	35.20
	Task4	78.91	52.59	55.49	48.97	69.53	49.90	98.53	86.17	-	-	-	-	59.41	23.25
	Task5	25.00	36.08	44.97	50.81	71.24	52.57	33.41	45.57	98.33	98.75	-	-	56.76	49.40
	Task6	19.73	46.54	42.40	49.85	34.85	41.29	3.37	50.09	28.49	50.00	99.76	99.84	56.27	68.34
Replay (Rondom)	Task1	85.06	85.30	-	-	-	-	-	-	-	-	-	-	85.30	-
	Task2	83.69	80.56	82.77	83.58	-	-	-	-	-	-	-	-	82.07	1.37
	Task3	86.04	77.50	63.38	64.55	99.59	99.35	-	-	-	-	-	-	80.47	9.69
	Task4	86.62	74.16	66.84	66.39	88.58	82.68	97.90	72.34	-	-	-	-	73.89	8.98
	Task5	85.45	65.84	60.33	53.72	95.81	94.64	95.62	50.26	99.29	98.90	-	-	72.67	7.42
	Task6	84.96	66.37	56.94	50.44	93.43	90.42	96.70	97.25	71.20	51.07	99.98	99.97	75.92	12.59
iCaRL (CVPR 2017)	Task1	85.06	85.30	-	-	-	-	-	-	-	-	-	-	85.30	-
	Task2	83.11	69.91	82.74	81.07	-	-	-	-	-	-	-	-	75.49	1.95
	Task3	60.94	74.88	52.78	58.77	98.54	97.58	-	-	-	-	-	-	77.08	27.04
	Task4	86.52	84.37	71.46	71.78	97.02	97.46	95.05	95.70	-	-	-	-	87.33	4.27
	Task5	87.50	89.04	62.79	66.31	97.56	97.37	86.78	88.38	98.79	98.96	-	-	88.01	7.30
	Task6	66.21	77.05	52.71	58.28	50.38	64.44	62.07	80.41	44.40	61.10	97.42	98.27	73.26	37.37
LwF (TPAMI 2018)	Task1	85.25	85.42	-	-	-	-	-	-	-	-	-	-	85.42	-
	Task2	54.69	70.75	52.71	58.65	-	-	-	-	-	-	-	-	64.70	30.56
	Task3	72.07	68.64	59.66	60.95	90.15	92.87	-	-	-	-	-	-	74.15	6.59
	Task4	78.81	50.00	57.49	50.04	69.78	50.00	97.32	60.29	-	-	-	-	52.58	9.66
	Task5	25.78	36.74	46.55	51.95	70.51	55.47	30.85	46.42	99.31	99.23	-	-	57.96	39.67
	Task6	21.19	46.63	43.31	50.28	31.06	38.60	3.54	50.18	30.09	51.12	99.95	99.95	56.13	60.50
PASS (CVPR 2021)	Task1	90.33	83.93	-	-	-	-	-	-	-	-	-	-	83.93	-
	Task2	79.88	75.95	62.23	63.34	-	-	-	-	-	-	-	-	69.65	10.45
	Task3	71.58	69.34	52.56	54.43	58.05	45.84	-	-	-	-	-	-	56.54	14.21
	Task4	37.01	50.10	44.08	48.85	52.67	49.06	40.86	44.63	-	-	-	-	48.16	25.62
	Task5	74.22	54.67	59.22	53.01	52.19	49.58	84.63	64.18	71.42	50.13	-	-	54.31	6.25
	Task6	79.20	62.55	59.39	54.52	41.41	53.72	83.94	72.54	71.14	50.37	56.52	50.68	57.40	6.32
ACIL (NeurIPS 2022)	Task1	85.55	76.85	-	-	-	-	-	-	-	-	-	-	85.55	-
	Task2	83.01	72.54	59.87	58.99	-	-	-	-	-	-	-	-	65.77	2.54
	Task3	81.05	68.61	61.53	60.38	80.93	74.51	-	-	-	-	-	-	67.83	2.25
	Task4	84.96	68.22	63.19	61.79	81.14	74.00	95.33	50.11	-	-	-	-	63.53	0.20
	Task5	85.45	69.54	63.68	62.10	80.60	72.90	95.54	50.22	69.44	51.90	-	-	61.33	0.16
	Task6	82.23	61.43	63.55	61.25	78.71	69.06	96.63	50.35	71.26	50.20	75.07	50.10	57.07	1.18
Fetril (WACV 2023)	Task1	86.04	85.75	-	-	-	-	-	-	-	-	-	-	85.75	-
	Task2	82.42	69.64	64.48	63.85	-	-	-	-	-	-	-	-	66.75	3.62
	Task3	70.90	54.41	53.91	51.52	78.03	66.65	-	-	-	-	-	-	57.53	12.86
	Task4	78.81	50.00	57.47	50.00	69.81	50.04	96.77	49.99	-	-	-	-	50.12	7.49
	Task5	53.12	36.57	52.57	47.12	61.39	44.75	87.64	51.37	71.34	50.20	-	-	46.36	17.66
	Task6	78.03	49.50	57.39	49.94	69.78	50.00	96.77	49.99	71.38	49.95	75.24	49.99	49.90	4.68
DFIL (MM 2023)	Task1	84.38	63.81	-	-	-	-	-	-	-	-	-	-	63.81	-
	Task2	84.96	72.77	82.70	80.50	-	-	-	-	-	-	-	-	76.64	0.00
	Task3	85.64	66.97	67.38	64.50	99.62	99.50	-	-	-	-	-	-	76.99	7.66
	Task4	85.55	69.10	65.34	61.01	94.72	91.75	97.75	66.17	-	-	-	-	72.01	7.45
	Task5	85.06	77.55	62.31	63.58	92.42	94.57	91.71	76.55	99.00	99.20	-	-	82.29	8.55
	Task6	84.86	79.45	57.30	58.00	95.18	93.66	95.54	97.69	61.95	47.03	99.81	99.62	79.24	13.98
DS-AL (AAAI 2024)	Task1	85.55	76.85	-	-	-	-	-	-	-	-	-	-	76.85	-
	Task2	81.64	71.00	59.36	58.45	-	-	-	-	-	-	-	-	64.73	3.91
	Task3	80.27	67.61	61.22	60.17	81.63	75.44	-	-	-	-	-	-	67.74	2.64
	Task4	84.18	67.39	62.73	61.30	81.95	74.91	95.19	49.17	-	-	-	-	63.19	0.46
	Task5	83.98	67.43	63.05	61.32	81.25	73.70	94.96	51.23	68.28	52.25	-	-	61.19	0.63
	Task6	82.23	61.10	63.15	60.69	78.98	69.43	96.43	49.81	71.23	50.42	74.88	50.11	56.93	1.26
MMAL (MM 2024)	Task1	89.75	84.23	-	-	-	-	-	-	-	-	-	-	84.23	-
	Task2	79.79	68.31	62.27	61.29	-	-	-	-	-	-	-	-	64.80	9.96
	Task3	78.61	65.54	61.62	60.44	80.90	74.44	-	-	-	-	-	-	66.81	5.90
	Task4	82.81	65.34	62.69	60.54	80.49	73.66	95.07	50.41	-	-	-	-	62.49	2.45
	Task5	82.81	66.35	62.42	60.35	80.33	72.74	92.43	52.09	69.47	55.45	-	-	61.40	2.60
	Task6	82.13	61.71	60.96	56.98	77.00	66.01	94.47	50.54	70.82	52.02	74.19	50.63	56.32	2.77
TagFex (CVPR 2025)	Task1	89.45	79.33	-	-	-	-	-	-	-	-	-	-	79.33	-
	Task2	85.06	65.93	76.73	72.99	-	-	-	-	-	-	-	-	69.46	4.39
	Task3	83.30	62.45	74.43	72.42	99.13	98.64	-	-	-	-	-	-	77.84	4.23
	Task4	86.52	69.55	66.48	62.46	98.16	98.22	97.18	61.52	-	-	-	-	72.94	4.72
	Task5	57.62	64.86	62.75	66.49	49.30	63.39	69.84	65.69	83.92	88.74	-	-	69.83	30.75
	Task6	41.80	62.23	49.24	55.62	92.59	94.18	57.60	78.10	46.44	62.28	99.89	99.90	75.39	31.75
Ours	Task1	91.37	84.03	-	-	-	-	-	-	-	-	-	-	91.37	-
	Task2	83.01	77.15	86.06	85.47	-	-	-	-	-	-	-	-	84.53	8.37
	Task3	69.79	63.82	59.79	58.88	98.51	97.56	-	-	-	-	-	-	76.03	23.93
	Task4	69.43	75.33	58.87	62.59	87.57	89.89	98.26	86.91	-	-	-	-	78.53	20.03
	Task5	75.55	70.45	60.40	58.70	92.33	90.95	89.30	86.20	98.66	98.26	-	-	83.25	14.15
	Task6	75.59	66.54	57.39	54.20	92.86	90.06	93.45	95.89	73.32	68.76	99.71	99.68	82.05	16.05

Table 7: Performance comparison of different methods on various datasets reverse

Method	Datasets	Task6		Task5		Task4		Task3		Task2		Task1		AAUC ↑	Fk ↓
		ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC		
Finetune	Task6	98.85	98.90	-	-	-	-	-	-	-	-	-	-	98.85	-
	Task5	26.44	50.13	94.54	93.66	-	-	-	-	-	-	-	-	60.40	72.41
	Task4	75.27	50.00	71.51	50.00	96.80	50.00	-	-	-	-	-	-	81.20	23.31
	Task3	22.86	17.97	68.16	49.67	61.20	34.66	97.08	95.17	-	-	-	-	62.33	45.99
	Task2	72.98	50.48	71.67	50.52	91.33	51.96	50.16	54.85	75.46	75.74	-	-	72.32	25.28
	Task1	52.57	50.67	64.59	51.87	65.96	64.56	43.18	52.92	61.13	64.07	91.41	82.59	63.13	35.06
Replay (Rondom)	Task6	98.85	98.90	-	-	-	-	-	-	-	-	-	-	98.85	-
	Task5	89.89	92.58	97.08	96.71	-	-	-	-	-	-	-	-	93.49	8.96
	Task4	87.80	76.69	76.69	61.91	96.89	74.87	-	-	-	-	-	-	87.13	15.72
	Task3	93.70	87.90	82.34	82.21	73.68	75.95	98.51	98.10	-	-	-	-	87.06	14.37
	Task2	81.19	62.68	73.91	55.25	91.36	59.82	80.90	70.25	81.63	80.23	-	-	81.80	15.99
	Task1	56.45	60.96	67.18	64.94	79.55	72.45	79.55	66.25	70.77	69.75	90.14	78.75	73.93	23.89
iCaRL (CVPR 2017)	Task6	98.85	98.90	-	-	-	-	-	-	-	-	-	-	98.85	-
	Task5	26.49	50.21	94.58	93.73	-	-	-	-	-	-	-	-	60.53	72.36
	Task4	76.02	51.52	72.63	52.88	96.86	50.90	-	-	-	-	-	-	81.84	22.39
	Task3	80.98	85.75	84.99	88.75	77.48	86.19	97.00	96.96	-	-	-	-	85.11	15.61
	Task2	47.76	59.35	76.17	77.23	74.11	78.79	72.94	79.32	78.62	77.73	-	-	69.92	29.08
	Task1	31.37	53.80	64.46	74.66	79.06	89.18	91.15	89.78	67.76	70.27	88.48	87.30	70.38	26.42
LwF (TPAMI 2018)	Task6	98.85	98.90	-	-	-	-	-	-	-	-	-	-	98.86	-
	Task5	28.98	52.81	45.02	61.54	-	-	-	-	-	-	-	-	37.00	69.87
	Task4	58.68	60.77	73.35	65.58	52.16	65.71	-	-	-	-	-	-	61.40	20.08
	Task3	50.85	36.56	69.11	50.66	75.00	44.84	90.85	84.95	-	-	-	-	71.45	17.41
	Task2	74.92	50.08	71.54	50.05	96.14	49.66	52.54	52.47	69.53	65.39	-	-	72.93	16.04
	Task1	60.34	50.36	69.31	50.07	77.82	61.54	42.78	56.81	57.64	60.89	91.11	82.06	66.50	24.17
PASS (CVPR 2021)	Task6	98.85	98.90	-	-	-	-	-	-	-	-	-	-	98.85	-
	Task5	24.93	50.06	87.41	91.05	-	-	-	-	-	-	-	-	56.17	73.92
	Task4	25.49	50.06	28.48	49.98	5.58	50.91	-	-	-	-	-	-	19.92	66.15
	Task3	67.49	49.26	12.47	15.87	92.67	50.04	73.70	56.48	-	-	-	-	61.59	35.43
	Task2	62.22	48.62	13.22	21.26	86.24	51.51	74.54	58.04	50.06	46.45	-	-	57.26	29.31
	Task1	27.02	49.03	28.22	49.53	15.17	48.78	67.12	60.15	42.69	46.32	34.86	49.58	35.85	44.66
ACIL NeurIPS 2022)	Task6	99.57	99.39	-	-	-	-	-	-	-	-	-	-	99.57	-
	Task5	99.29	98.76	83.76	88.15	-	-	-	-	-	-	-	-	91.52	0.28
	Task4	99.06	98.28	83.93	88.02	53.60	71.24	-	-	-	-	-	-	78.86	0.25
	Task3	98.89	97.95	83.93	87.96	54.44	72.11	85.17	76.92	-	-	-	-	80.61	0.23
	Task2	98.76	97.68	83.96	87.80	54.81	71.43	85.36	77.08	50.86	49.78	-	-	74.75	0.20
	Task1	98.71	97.57	83.88	87.68	54.98	71.96	85.36	76.90	51.94	49.55	74.71	52.45	74.93	0.19
Fetril (WACV 2023)	Task6	98.85	98.90	-	-	-	-	-	-	-	-	-	-	98.85	-
	Task5	92.73	87.62	84.49	83.51	-	-	-	-	-	-	-	-	88.61	6.12
	Task4	75.27	50.00	74.40	49.93	96.80	50.00	-	-	-	-	-	-	81.16	16.83
	Task3	23.64	26.21	67.07	57.62	55.24	45.08	85.82	77.36	-	-	-	-	57.94	44.73
	Task2	23.50	15.70	64.41	45.10	85.02	44.35	73.24	56.03	57.26	50.50	-	-	60.69	29.95
	Task1	19.99	13.36	62.27	43.57	83.41	43.08	71.24	52.44	57.47	50.00	78.81	50.00	62.20	25.81
DFIL (MM 2023)	Task6	98.85	98.90	-	-	-	-	-	-	-	-	-	-	98.85	-
	Task5	27.28	50.49	91.32	90.67	-	-	-	-	-	-	-	-	59.30	71.57
	Task4	67.63	55.59	71.04	60.77	96.17	60.56	-	-	-	-	-	-	78.28	25.75
	Task3	84.93	70.48	76.83	66.73	90.70	57.74	99.05	98.84	-	-	-	-	87.88	11.29
	Task2	83.15	68.99	75.12	61.42	89.69	66.80	81.93	73.83	78.14	77.95	-	-	81.60	13.88
	Task1	39.99	52.74	65.99	66.92	77.30	73.03	90.45	87.09	69.76	70.76	91.50	83.15	72.50	24.01
DS-AL (AAAI 2024)	Task6	99.58	99.41	-	-	-	-	-	-	-	-	-	-	99.58	-
	Task5	99.02	98.21	85.33	89.05	-	-	-	-	-	-	-	-	92.18	0.56
	Task4	98.39	96.91	85.18	87.88	60.57	70.05	-	-	-	-	-	-	81.38	0.67
	Task3	98.07	96.27	85.23	87.82	60.60	69.63	86.01	77.87	-	-	-	-	82.48	0.54
	Task2	97.69	95.48	85.05	87.21	61.49	68.79	85.80	77.64	54.21	51.27	-	-	76.85	0.60
	Task1	97.58	95.27	84.90	86.94	61.58	68.83	85.88	77.55	54.25	50.00	76.37	50.81	76.76	0.51
MMAL (MM 2024)	Task6	99.58	99.43	-	-	-	-	-	-	-	-	-	-	99.58	-
	Task5	97.63	95.40	86.46	88.27	-	-	-	-	-	-	-	-	92.05	1.95
	Task4	90.60	80.69	80.93	74.16	90.18	56.60	-	-	-	-	-	-	87.16	7.25
	Task3	89.90	79.83	80.77	73.65	89.78	56.39	86.58	78.64	-	-	-	-	86.76	5.26
	Task2	88.06	76.09	79.70	70.52	90.32	56.67	86.36	78.53	55.85	50.80	-	-	80.06	4.62
	Task1	87.87	75.69	79.42	70.01	90.29	56.66	86.34	78.41	56.52	50.42	78.81	51.52	79.88	3.80
TagFex (CVPR 2025)	Task6	99.28	99.25	-	-	-	-	-	-	-	-	-	-	99.28	-
	Task5	91.78	83.45	84.09	80.39	-	-	-	-	-	-	-	-	87.93	7.50
	Task4	75.27	50.00	71.51	50.00	96.80	50.00	-	-	-	-	-	-	81.20	25.89
	Task3	74.65	50.24	71.15	50.24	95.82	51.24	47.05	61.81	-	-	-	-	72.17	13.89
	Task2	69.08	58.53	73.31	55.69	77.82	61.98	62.82	64.13	71.46	70.30	-	-	70.90	22.57
	Task1	45.88	50.90	62.61	57.06	70.13	63.23	80.95	69.45	60.98	61.94	87.79	76.25	68.06	29.92
Ours	Task6	99.33	99.27	-	-	-	-	-	-	-	-	-	-	99.33	-
	Task5	98.44	97.51	95.92	94.35	-	-	-	-	-	-	-	-	87.93	0.89
	Task4	86.06	69.83	77.20	62.24	97.09	60.17	-	-	-	-	-	-	81.20	0.45
	Task3	90.32	93.16	79.50	85.39	47.12	71.82	96.46	94.31	-	-	-	-	72.17	10.66
	Task2	75.98	81.98	76.12	79.66	39.98	64.21	80.09	81.98	80.60	80.76	-	-	10.66	70.90
	Task1	49.09	54.40	62.50	54.69	70.07	66.68	90.64	90.27	70.12	71.63	90.72	81.31	68.06	23.33

Table 8: Benchmark results under the cross-technology incremental setting in CIL. The best results are highlighted in red, and the sub-optimal results are underlined.

Method	Class 1		Class 2		Class 3		Class 4		Class 5		Class 6		AIA ↑	AF ↓
	AA	BWT	AA	BWT	AA	BWT	AA	BWT	AA	BWT	AA	BWT		
Finetune	89.29	-	68.06	-21.23	81.25	-8.04	65.56	-23.73	24.32	-64.97	26.77	-62.51	59.21	30.08
Replay (Rondom)	86.44	-	60.42	-26.02	71.09	-15.35	60.07	-26.37	30.61	-55.83	35.33	-51.11	57.32	29.12
iCaRL (CVPR 2017)	86.44	-	64.43	-22.01	75.20	-11.24	56.59	-29.85	35.69	-50.75	42.10	-44.34	60.08	26.36
Lwf (TPAMI 2018)	83.93	-	76.62	-7.31	83.33	-0.60	68.10	-15.83	25.39	-58.54	27.05	-56.88	60.74	23.19
PASS (CVPR 2021)	77.16	-	68.06	-9.10	60.12	-17.03	41.16	-35.99	47.25	-29.90	31.16	-45.99	54.15	23.01
ACIL (NeurIPS 2022)	87.95	-	81.94	-6.00	67.82	-20.12	68.30	-19.64	66.68	-21.27	62.63	-25.32	72.55	15.40
Fetril (WACV 2023)	85.71	-	66.67	-19.06	61.19	-24.53	58.10	-27.61	24.02	-61.69	24.39	-61.32	53.35	32.36
DFIL (MM 2023)	93.75	-	52.08	-41.67	24.54	-69.21	13.22	-80.53	13.60	-80.15	12.42	-81.33	34.94	58.82
DS-AL (AAAI 2024)	87.95	-	81.94	-6.00	68.06	-19.89	68.61	-19.33	66.85	-21.09	62.73	-25.22	72.69	15.26
MMAL (MM 2024)	88.84	-	73.84	-15.00	64.89	-23.95	66.14	-22.70	65.67	-23.17	55.79	-33.04	69.20	19.65
TagFex (CVPR 2025)	91.07	-	78.47	-12.60	85.03	-6.04	78.32	-12.75	72.12	-18.95	68.82	-20.26	78.97	12.10
Ours	94.35	-	76.47	-17.88	87.24	-7.10	76.27	-18.07	77.90	-16.44	63.26	-31.09	79.25	15.10

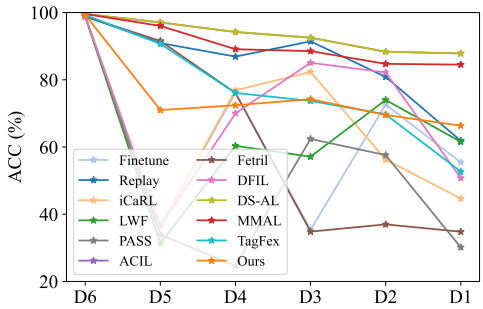

Figure 7: The ACC (%) of continual detection performance in domain incremental learning following a reverse sequence.

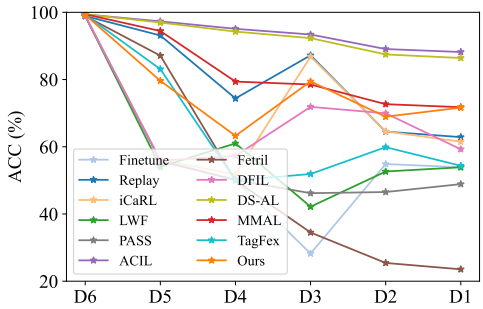

Figure 8: The AUC (%) of continual detection performance in domain incremental learning following a reverse sequence.

Table 9: Detailed performance in the modality incremental setting.

Method	Datasets	Video-only		Audio-only		Video-Audio		AA ↑	AAUC ↓	Fk ↓
		ACC	AUC	ACC	AUC	ACC	AUC			
Finetune	Video-only	93.35	88.38	-	-	-	-	93.35	88.38	-
	Audio-only	59.70	67.74	94.86	91.52	-	-	77.28	79.63	33.65
	Video-Audio	41.30	58.45	46.76	61.99	54.93	68.34	47.66	62.93	50.08
Repaly (Rondom)	Video-only	93.32	88.38	-	-	-	-	93.32	88.38	-
	Audio-only	87.97	89.64	94.93	94.34	-	-	91.45	91.99	5.35
	Video-Audio	85.58	88.30	88.45	90.45	84.94	88.25	86.33	89.00	7.11
iCaRL (CVPR 2017)	Video-only	93.32	88.31	-	-	-	-	93.32	88.31	-
	Audio-only	87.63	85.04	92.53	87.76	-	-	90.08	86.40	5.69
	Video-Audio	81.43	86.30	88.61	91.29	92.39	93.99	87.48	90.53	7.90
LwF (TPAMI 2018)	Video-only	93.37	88.44	-	-	-	-	93.37	88.44	-
	Audio-only	80.75	78.32	90.32	84.11	-	-	85.54	81.22	12.62
	Video-Audio	67.50	77.26	92.30	94.35	83.94	88.76	81.25	86.79	12.94
PASS (CVPR 2021)	Video-only	93.33	88.46	-	-	-	-	93.33	88.46	-
	Audio-only	58.51	67.03	94.78	91.34	-	-	76.64	79.19	34.82
	Video-Audio	34.95	55.44	69.59	78.97	36.35	56.97	46.96	63.79	41.79
ACIL (NeurIPS 2022)	Video-only	94.07	90.62	-	-	-	-	94.07	90.62	-
	Audio-only	94.35	90.61	91.88	88.52	-	-	93.12	89.57	0
	Video-Audio	77.80	64.79	76.22	64.40	79.17	65.67	77.73	64.95	16.11
Fetril (WACV 2023)	Video-only	93.30	88.31	-	-	-	-	93.30	88.31	-
	Audio-only	88.17	87.03	91.77	89.22	-	-	89.97	88.13	5.13
	Video-Audio	67.47	53.74	62.80	51.74	76.46	59.19	68.91	54.89	27.40
DFIL (MM 2023)	Video-only	93.73	90.01	-	-	-	-	93.73	90.01	-
	Audio-only	92.13	90.07	94.93	91.79	-	-	93.53	90.93	1.60
	Video-Audio	89.27	90.97	90.43	92.42	97.64	97.08	92.45	93.49	4.48
DS-AL (AAAI 2024)	Video-only	94.08	90.73	-	-	-	-	94.08	90.73	-
	Audio-only	94.22	90.40	92.68	89.25	-	-	93.45	89.83	0
	Video-Audio	73.95	61.41	72.48	61.52	79.53	63.35	75.32	62.09	20.24
MMAL (MM 2024)	Video-only	93.92	90.44	-	-	-	-	93.92	90.44	-
	Audio-only	93.80	89.94	92.07	88.22	-	-	92.94	89.08	0.12
	Video-Audio	65.70	51.75	59.87	48.46	77.17	58.62	67.58	52.94	30.21
TagFex (CVPR 2025)	Video-only	94.12	91.00	-	-	-	-	94.12	91.00	-
	Audio-only	91.85	90.00	95.92	92.82	-	-	93.89	91.41	2.27
	Video-Audio	82.17	83.37	87.24	87.34	95.46	92.45	88.30	87.72	10.32
Ours	Video-only	96.80	97.40	-	-	-	-	96.80	93.28	-
	Audio-only	79.88	86.21	93.78	95.46	-	-	86.83	90.83	16.93
	Video-Audio	91.99	89.20	93.10	89.85	93.84	90.62	92.98	89.89	2.74

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